

PACKPROOF

Smart Packaging Verification System

Master Project Documentation & Handover

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1. Project Overview

Project Name: PackProof

Project Status: Partially Completed (PCB design pending)

Platform: Raspberry Pi Zero 2W

Target Environment: Factory outlets and packaging units

PackProof is a smart factory-side verification system designed to record and validate uniform product packaging. The system captures continuous video evidence of the packaging process against a user-entered Order ID / Invoice ID and uploads it to a central server for traceability and audit purposes.

The system is designed with a strong focus on:

- Power-failure safety
- Offline operation
- Automatic recovery
- Minimal operator interaction

2. Problem Statement & Solution

2.1 The Problem

In factory environments, it is difficult to prove whether products were packed correctly and uniformly, especially when disputes arise later. There is no reliable audit trail linking the packaging process to the final product.

2.2 The Solution

PackProof ensures:

- Every package is recorded on video
- Each video is linked to an Order ID / Invoice ID
- Videos are uploaded automatically to a server
- Power loss does not corrupt recordings

- No technical knowledge is required by factory staff
- Automatic recovery from connection failures

This creates a complete, tamper-proof audit trail for dispute resolution and quality assurance.

3. System Architecture (High-Level)

The system consists of the following logical modules:

- Touchscreen-based Kiosk UI (Operator interface)
- Camera-based video recorder (Continuous capture)
- Background uploader service (Automatic file transmission)
- Power-loss detection & safe shutdown module (Safety)
- Automatic reboot & power isolation circuit (Hardware control)

All modules operate automatically once the system is powered ON. No manual intervention is required during normal operation.

4. Hardware Configuration

Component	Specification
Controller	Raspberry Pi Zero 2W
Camera	Raspberry Pi Camera Module
Display	3.5-inch Touchscreen Display
Power Input	5V Adapter
UPS Module	Battery-backed UPS with mains detect signal
Power Control Circuit	Custom relay + transistor-based auto-ON circuit
Storage	microSD card (32GB recommended)
PCB Status	Zero PCB prototype completed Final PCB design pending

5. Software Stack

Operating System: Raspberry Pi OS (Linux-based)

Programming Language: Python 3

UI Framework: Tkinter + ttkbootstrap

Camera Framework: Picamera2 + FFmpeg

Service Manager: systemd

Wi-Fi Configuration: nmcli-based management

Server Communication: HTTPS with certificate validation

6. Boot & Startup Flow

On Power-ON:

1. Raspberry Pi boots automatically when adapter power is available
2. Operating system loads normally
3. systemd launcher service starts automatically
4. app.py launches the kiosk application
5. System checks internet connectivity:
 - If connected → Recorder and uploader services start immediately
 - If not connected → Wi-Fi setup interface appears
6. Background uploader service runs continuously
7. Power monitor service runs in parallel

7. Functional Working Flow

7.1 Operator Interaction Flow

1. Operator enters Order ID / Invoice ID on the touchscreen
2. Live camera preview is displayed
3. Operator taps "Start Recording" button
4. Packaging process is recorded continuously
5. Operator taps "Stop Recording" after completion
6. Video is automatically saved with Order ID as filename

7.2 Data Handling & Storage

- Video is saved locally with naming format: [InvoiceID]_[Timestamp].mp4
- Invoice image is also captured and stored(optional)
- Entry is added to local upload queue with metadata
- Videos remain stored until confirmed uploaded

7.3 Background Upload Service

- Upload happens automatically in the background
- If internet is unavailable, files remain queued
- Upload resumes automatically once connectivity returns
- Successfully uploaded files are deleted from local storage
- Failed uploads are retried with exponential backoff

8. Power Failure Handling & Safety Design

This is a CRITICAL design feature that ensures zero data loss during unexpected power failures.

8.1 Power Loss Detection Mechanism

- UPS module provides a mains-detect signal
- GPIO26 monitors adapter power presence via resistor divider circuit
- GPIO input is fully protected from over-voltage using voltage divider
- Detection occurs within milliseconds of power loss

8.2 Safe Shutdown Logic on Power Loss

When power loss is detected:

1. Fullscreen warning popup appears immediately
2. Video recording is safely stopped
3. File system is synced to prevent data corruption
4. Upload queue is preserved in SQLite database
5. After 5-10 second timeout, system performs clean shutdown

8.3 Protection Against

- Video file corruption
- SD card damage from sudden power-off

- File system errors (fsck repairs avoided)
- Data loss of recorded metadata
- Battery drain when device is off

9. Automatic Power Control Circuit

A relay + transistor-based auto power control circuit was designed and implemented to solve a critical real-world problem: eliminating manual power control while ensuring safe operation.

Achievements:

- Raspberry Pi auto-boots when adapter power is available (no manual button needed)
- Pi continues running on battery during graceful shutdown
- Pi is fully disconnected after shutdown (power isolation)
- Battery does not drain overnight
- System restarts automatically when power returns
- Factory staff requires zero technical knowledge

This design ensures long-term unattended operation suitable for factory environments where manual intervention is impractical.

10. Code Organization

Full source code is intentionally excluded from this document as per best handover practices.

High-level code structure:

- launcher.service: Systemd service for auto-start
- app.py: Main kiosk application UI
- recorder.py: Video capture and frame handling
- uploader.py: Background file transmission with retry logic
- power_monitor.py: GPIO power monitoring and safe shutdown
- wifi.py: Wi-Fi setup interface

All services are managed using systemd for automatic restart on failure.

11. Current Status & Completion Summary

Item	Status
Complete system design and architecture	Completed
Power-safe operation with UPS integration	Completed
Automatic power control circuit design	Completed
Touchscreen kiosk UI	Completed
Background uploading with queue management	Completed
Offline operation support	Completed
Field-ready functional prototype	Completed
Final schematic capture (from zero PCB)	Completed

12. Handover Notes

- System is stable and field-ready for factory deployment
- Operates safely in environments with unreliable power
- Suitable for continuous 24/7 operation
- PCB design can be completed using the existing zero-PCB circuit
- No dependency on original developer for day-to-day operation
- Minimal training required for factory staff (touchscreen only)
- All configurations are documented and accessible

13. Final Summary

PackProof is a robust, power-safe, factory-grade packaging verification system designed with real-world constraints in mind.

The project demonstrates strong expertise in:

- Embedded systems software design (Python + systemd)
- Hardware power management and circuit design
- Real-world reliability engineering
- Offline-first application architecture
- Factory environment adaptability

The system is ready for immediate deployment pending final PCB fabrication.